

ADVANCES IN HEART FAILURE, MECHANICAL CIRCULATORY SUPPORT AND TRANSPLANT

Kidney Failure in Cardiogenic Shock

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ABSTRACT: Acute kidney injury (AKI) is a frequent and prognostically decisive complication of cardiogenic shock (CS), conferring substantially increased short- and long-term mortality. As understanding of CS has evolved beyond an isolated low-output model, AKI is increasingly recognized as a consequence of multifactorial pathophysiology encompassing venous congestion, neurohormonal dysregulation, microcirculatory dysfunction, inflammation, and iatrogenic injury from shock therapies. Despite this expanded mechanistic framework, approaches to the diagnosis and management of AKI in CS remain highly variable, and no single strategy has consistently improved renal or survival outcomes. Emerging data suggest multiple biologically distinct phenotypes of AKI in CS that may enable future targeted therapeutics. This review synthesizes contemporary evidence on the classification, epidemiology, and mechanisms of AKI in CS across distinct phenotypes, including early- versus late-onset, acute versus acute-on-chronic, and cardiorenal versus cardiometabolic presentations. We further discuss the renal effects of contemporary shock therapies, including vasoactive agents, temporary mechanical circulatory support, and renal replacement strategies. Finally, we highlight emerging diagnostic tools, including noninvasive vascular imaging and renal biomarkers, alongside novel device-based approaches to improve renal perfusion and congestion. Advancing care in CS will depend on redefining AKI not only as a binary marker of renal dysfunction but also as a mechanistically heterogeneous component of shock biology amenable to targeted intervention.

Key Words: biology ■ hyperemia ■ inflammation ■ microcirculation ■ perfusion

Acute kidney injury (AKI) occurs frequently in cardiogenic shock (CS) in up to one-third of patients,¹ defining the Type I cardiorenal syndrome (CRS). It is consistently and strongly associated with an increased risk of in-hospital and posthospital mortality,^{2–4} although the direct pathophysiologic derangements of AKI, such as hyperkalemia, acidosis, and hypervolemia, can be corrected with diuretics or renal replacement therapy (RRT). Furthermore, there is a strong correlation between CS mortality and the intensity of therapies needed to overcome renal deficiencies.⁵ This confirms that the downstream effects of AKI are not the primary cause of mortality per se, but rather that CRS is an integral component of the CS pathophysiologic syndrome or that the severity of CRS is a bellwether for critical upstream pathophysiologic mechanisms that are underrecognized and undertreated.

Why does this relationship exist between CRS and CS mortality? Early ideas that CRS is simply a manifestation of deficient cardiac output have been challenged: glomerular filtration can be maintained across a wide range of renal blood flow,⁶ and no study has demonstrated a meaningful correlation between lower cardiac output and AKI.^{4,7} Instead, there is growing evidence that CRS represents a complex interplay of cardiac, renal, neurohormonal, and inflammatory derangements. CRS mechanisms in CS may be worse versions of the same CRS mechanisms observed in acute decompensated heart failure (ADHF), but some mechanisms may differ in substantive ways. Moreover, traditional CS therapies have shown themselves inadequate to completely reverse this systemic pathophysiology, or even potentially harmful. Innovative therapies inspired by these concepts are being developed to ameliorate CRS pathophysiology and, hopefully, CS outcomes overall.

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Nonstandard Abbreviations and Acronyms

ADHF	acute decompensated heart failure
AKI	acute kidney injury
AKIKI	Artificial Kidney Initiation in Kidney Injury Trial
AMI-CS	acute myocardial infarction related cardiogenic shock
Ang II	angiotensin II
CARESS-HF	Cardiorenal Rescue Study in Acute Decompensated Heart Failure
CICU	cardiac intensive care unit
CKD	chronic kidney disease
CRRT	continuous renal replacement therapy
CRS	cardiorenal syndrome
CS	cardiogenic shock
CVP	central venous pressure
DOREMI	Milrinone as Compared with Dobutamine in the Treatment of Cardiogenic Shock
DOSE	Diuretic Strategies in Patients with Acute Decompensated Heart Failure
DPP3	dipeptidyl peptidase III
ECMO	extracorporeal membrane oxygenation
ELAIN	Early versus Late Initiation of Renal Replacement Therapy in Critically Ill Patients with Acute Kidney Injury
HF-CS	heart failure related cardiogenic shock
IGFBP7	insulin-like growth factor-binding protein 7
IHD	intermittent hemodialysis
KDIGO	Kidney Disease Improving Global Outcomes
KIM-1	kidney injury marker -1
L-FABP	liver-type fatty acid-binding protein
NGAL	neutrophil gelatinase-associated lipocalin
RAAS	renin angiotensin aldosterone system
RPP	renal perfusion pressure
RRT	renal replacement therapy
SCAI	Society for Cardiovascular Angiography
STARRT-AKI	Standard versus Accelerated Initiation of Renal-Replacement Therapy in Acute Kidney Injury
TIMP-2	tissue inhibitor of metalloproteinases-2
tMCS	temporary mechanical circulatory support

DEFINITIONS AND CLASSIFICATIONS

Cardiogenic Shock

CS is a clinical syndrome triggered by impaired systemic perfusion due to cardiac insufficiency. CS definitions from landmark clinical trials include evidence of hypoperfusion (cool extremities, elevated lactate, end-organ damage) with sustained systolic blood pressure (SBP) <90 mmHg or requiring vasopressors, inotropes, or temporary mechanical circulatory support (tMCS).^{8–10} Normotensive CS has also been recognized, with malperfusion and reduced cardiac output despite preserved blood pressure. In 2019, the Society for Cardiovascular Angiography and Interventions (SCAI) introduced a staging system for CS severity.¹¹ SCAI stages predict in-hospital and long-term mortality and have been validated in diverse CS populations.¹² In addition, the severity of renal dysfunction increases incrementally with higher SCAI stages, with the incidence of severe AKI and RRT highest in patients with stages D (Deteriorating) and E (Extremis) shock.¹³

CS Phenotypes

CS can be divided into different etiologic, anatomic, and hemometabolic phenotypes, which carry varying prognoses. The Shock Academic Research Consortium criteria divide CS etiologies into acute myocardial infarction (AMI-CS), acute-on-chronic or de novo heart failure (HF-CS), postcardiotomy, or secondary related to arrhythmias, valve disease, or pericardial disease.¹⁴ Additional anatomic modifiers include the presence of predominantly left, right, or biventricular dysfunction. Hemodynamic modifiers include whether a patient is congested and vasoconstricted, vasodilatory or mixed, normotensive, or euvolemic, noting the increased recent attention to the high morbidity and mortality of the mixed vasodilated phenotype.¹⁵ Most evidence in CS comes from AMI-CS or ADHF, neither of which generalizes perfectly to HF-CS. Machine learning techniques have additionally clustered patients into phenotypes termed noncongested, cardiorenal, and cardiometabolic. The cardiometabolic phenotype exhibits more acute end-organ dysfunction with type 1 CRS, liver injury, lactic acidosis, and inflammation, and it has the highest mortality. The cardiorenal phenotype tends to be older patients with chronic kidney disease (CKD) and may represent more type 2 CRS, and mortality is in between that of the cardiometabolic and noncongested phenotypes.¹⁶

Classification of AKI in CS

Assessment of AKI is an important component of models of CS prognosis (Table 1). The best-validated definitions include AKI by Kidney Disease: Improving Global Outcomes (KDIGO) criteria¹⁷ or AKI requiring RRT. KDIGO

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Table 1. AKI Definitions in CS

	SCr/eGFR	Urine output	Timeframe	Comments
KDIGO				Most used standardized definition Variations used in trials of noncardiogenic shock Reliant on creatinine which can be affected by body composition, diet, common cardiac medications, volume of distribution Creatinine change can be in reference to pre- or within-hospital measurements Highly variable definitions of baseline creatinine depending on the study
Stage 1 AKI	Absolute increase ≥ 0.3 mg/dL SCr OR ≥ 1.5 – $2\times$ baseline	<0.5 mL/kg per h for 6–12 h	≤ 48 h	
Stage 2 AKI	≥ 2 – $3\times$ baseline SCr	<0.5 mL/kg per h for 12–24 h		
Stage 3 AKI	Absolute increase ≥ 0.5 mg/dL from baseline SCr to ≥ 4 mg/dL OR $\geq 3\times$ baseline OR initiation of RRT	<0.3 mL/kg per h for ≥ 24 h OR anuria for ≥ 12 h		
RIFLE				Largely replaced by KDIGO guidelines Similar criteria as KDIGO with identical drawbacks inherent to creatinine, including necessary knowledge of baseline and variations depending on individual and outside factors Additionally reliant on eGFR which does not accurately capture acute changes in renal function Inclusive of larger timeframe than KDIGO
Risk	KDIGO ≥ 1.5 – $2\times$ baseline SCr OR eGFR decrease $>25\%$	<0.5 mL/kg per h for 6–12 h	1–7 d	
Injury	≥ 2 – $3\times$ baseline SCr OR eGFR decrease $>50\%$	<0.5 mL/kg per h for 12–24 h		
Failure	$\geq 3\times$ baseline SCr OR SCr > 4 mg/dL OR eGFR decrease $>75\%$	<0.3 mL/kg per h for ≥ 24 h OR anuria for ≥ 12 h		
WRF				Recommended by European Heart Journal and Heart Failure Academic Research Consortium for assessment of kidney dysfunction in HF Reliant on creatinine with inherent flaws as above Uses in-hospital creatinine only Emphasizes trajectory of renal dysfunction and kidney response to CS management Highly variable timepoints depending on the study
	Absolute increase ≥ 0.3 mg/dL OR decrease of eGFR $>25\%$ OR initiation of RRT	NA	Variable	

AKI definitions vary in CS. The KDIGO classification system is the most used, largely replacing the older RIFLE criteria. WRF is a newer definition which accounts for early changes in creatinine during hospitalization. Each has strengths and limitations defining AKI in a cardiogenic shock population. AKI indicates acute kidney injury; CS, cardiogenic shock; eGFR, estimated glomerular filtration rate; HF, heart failure; KDIGO, Kidney Disease: Improving Global Outcomes; NA, not available; OR, odds ratio; RIFLE, Risk, Injury, Failure, Loss, End-Stage; RRT, renal replacement therapy; SCr, serum creatinine; and WRF, worsening renal function.

stage predicts CS survival, as patients with moderate-to-severe AKI ($\geq$ KDIGO Stage 2) have increased short- and long-term mortality.²

Despite being the most used and evidence-based AKI classification tool, numerous limitations exist in applying KDIGO to CS. First, in patients without preexisting CKD, significant changes in kidney function can be accommodated before a change in measured serum creatinine is observed. This is less so in patients with CKD, where the relationship between declining estimated glomerular filtration rate and serum creatinine becomes more linear. Yet, irrespective of baseline kidney function, there can be a substantial delay in the rise of creatinine when renal clearance is not in a steady state, and baseline prehospital creatinine levels may not be readily available for comparison. This can result in AKI misclassification in patients who instead have CKD, or delays in the diagnosis of AKI in CS when a rise in creatinine has not yet manifested. Early application of the urine output KDIGO criteria can overcome the delay in diagnosis if urine output can be accurately quantified. Nevertheless, decreased intrinsic urine output can be masked by diuretic use and can be equally confounded by CKD. Moreover, when estimated glomerular filtration rate decreases as a result of effective intravascular decongestion, this may reflect renal

hemodynamic effects rather than true injury, and some studies have shown this to be associated with improved survival.¹⁸ Finally, lower serum creatinine may reflect sarcopenia or hemodilution,¹⁹ while higher serum creatinine at admission may reflect misclassified CKD rather than AKI.

Worsening renal function has been proposed as an alternative definition of AKI in CS, incorporating a change between 2 separate measurements of creatinine early during a hospitalization. This aims to characterize kidney function trajectory rather than a single threshold event.^{20,21} However, similar to KDIGO criteria, worsening renal function remains fundamentally dependent on absolute or relative changes in creatinine and thus inherits many of the same limitations, and it has yet to be widely adopted.

EPIDEMIOLOGY AND PROGNOSTIC SIGNIFICANCE OF AKI IN CS

Incidence

Early studies in AMI-CS reported a high AKI incidence ranging from 35% to 55%,^{1,22} with 12% to 16% requiring RRT (Table 2).^{22,23,29} The incidence varied by

Table 2. Incidence and Mortality Association of AKI or RRT by CS Cause

References	Study design	N	AKI definition	Outcome	AKI		RRT	
					Incidence	Impact	Incidence	Impact
AMI-CS								
Vallabhajosyula 2019 ¹	Administrative database	440 257	ICD-9 codes	In-hospital mortality	35.3%	OR 1.67 (vs no AKI)	3.4%	OR, 2.16 (vs no AKI)
Lauridsen 2015 ²³	Administrative database	5079	ICD-10 codes	In-hospital mortality	NA	NA	13%	aOR, 1.70 (vs no RRT)
DanGer Shock: Moller 2024 ¹⁰ , Zweck 2024 ²⁴	RCT (Impella CP vs standard care)	360	KDIGO Stage 2 from 2x SCr ≥12 h apart	180 d all-cause mortality	61% (Impella CP)	54% vs 33% (no AKI)	42% (Impella CP)	56% vs 38% (no RRT)
					45% (Control)	68% vs 51% (no AKI)	27% (Control)	74% vs 53% (no RRT)
IABP-II (Thiele 2012) ⁹	RCT (IABP vs standard care)	600	RRT	NA	NA	NA	21% (IABP)	NA
					NA	NA	16% (Control)	NA
Kataria 2025 ⁴	Multiinstitutional cohort (CSWG)	1718	WRF within 72 h of shock	In-hospital mortality	41%	49% vs 29% (no WRF)	20.5%	63% (no comparison)
Varshney 2025 ²⁵	Multiinstitutional cohort (CCCTN)	1146	RRT	In-hospital mortality	NA	NA	13.3%	aOR, 3.42* (vs no severe AKI/RRT)
HF-CS								
Kataria 2025 ⁴	Multiinstitutional cohort (CSWG)		WRF within 72 h of shock	In-hospital mortality	32%	36% vs 18% (no WRF)	19.60%	47% (no comparison)
Varshney 2025 ²⁵	Multiinstitutional cohort (CCCTN)	1989 (acute on chronic HF)	RRT	In-hospital mortality	NA	NA	13.9%	aOR, 3.96* (vs no severe AKI/RRT)
		769 (de novo HF)			NA	NA	13.90%	aOR, 2.72* (vs no severe AKI/RRT)
Sundermeyer 2025 ²⁶	Multiinstitutional cohort	1010	KDIGO criteria at 24 h	30-d all-cause mortality	38.70%	NA	31.0%	OR, 1.41, <i>P</i> <0.001 (vs no RRT)
Postcardiotomy CS								
Rastan 2010 ²⁷	Single institution cohort	517	RRT	In-hospital mortality	NA	NA	65%	OR, 4.27 (vs no RRT)
Li 2024 ²⁸	Single institution cohort	216	KDIGO criteria	In-hospital mortality	84%	NA	53%	OR, 3.81 (vs no RRT)

The incidence of AKI or RRT and their associations with mortality differs by CS cause. This table reviews observational studies and randomized controlled trials that produced incidence and mortality estimates in AMI-CS, HF-CS, and postcardiotomy CS. AKI indicates acute kidney injury; AMI-CS, acute myocardial infarction cardiogenic shock; aOR, adjusted odds ratio; CCCTN, Critical Care Cardiology Trials Network; CS, cardiogenic shock; CSWG, Cardiogenic Shock Working Group; HF, heart failure; HF-CS, heart failure related cardiogenic shock; IABP, intra-aortic balloon pump; ICD, International Classification of Diseases; KDIGO, Kidney Disease: Improving Global Outcomes; NA, not available; OR, odds ratio; RCT, randomized controlled trial; RRT, renal replacement therapy; and WRF, worsening renal function.

*aOR for severe AKI including RRT and SCr ≥3.5 mg/dL.

definition, with 1 study reporting a 31% incidence based on creatinine versus 50% based on urine output.³⁰ Subsequently, the high AKI incidence has been confirmed across diverse CS etiologies, and it has increased over time. One study of 19 416 CS admissions found that 32% of patients to had AKI between 1997 and 2000 and 54% between 2009 and 2012.³¹ The incidence of severe AKI requiring RRT or with creatinine ≥3.5 mg/dL is similar across non-cardiac surgery CS etiologies, at 25% of admissions.²⁵ Postcardiotomy populations with CS have even higher rates of severe AKI. For example, a meta-analysis of 1866 cardiac surgery patients on extracorporeal membrane oxygenation (ECMO) showed 46% requiring RRT,³² whereas a lower incidence of severe AKI is observed in noncardiac surgery CS patients even when on ECMO or other tMCS.³³

Prognostic Significance

AKI strongly predicts inpatient mortality in all CS phenotypes, with an ≈50% to 150% increased mortality risk.^{1,5,23,34} The risk of death is proportional to the severity of AKI and is strongest when AKI occurs within 24 hours of cardiac intensive care unit (CICU) admission.² Patients with AKI receive more intensive CICU treatment, including increased vasoactive-inotropic scores and tMCS.^{4,26} The association of AKI and mortality persists long after hospitalization, with 5-year mortality among 3059 survivors of AMI-CS at 43% for those with AKI requiring RRT compared with 29% for those who did not.²³ Among this same cohort, patients with AKI-RRT had a 5-year risk of chronic dialysis of 11% versus 1% for patients with non-AKI-RRT.²³

MECHANISMS OF CARDIORENAL SYNDROME IN CS

The term CRS was first coined to conceptualize renal dysfunction in ADHF. This initial cardiocentric definition has since expanded to 5 categories based on acuity and primary organ dysfunction.³⁵ Although this expanded classification is intended to guide diagnostic and therapeutic strategies, in practice these phenotypes often overlap or transition and represent additional within-group mechanistic heterogeneity.

Acutely decreased arterial flow and increased venous congestion, which cause lower renal perfusion pressure (RPP), initiate type I CRS. Yet, nonhemodynamic factors such as neurohormonal activation, inflammation, microcirculatory dysfunction, and iatrogenic injury propagate CRS and help more fully explain its association with mortality (Figure 1).

Hemodynamic Mechanisms

CS begins with an ailing heart's inability to maintain adequate forward flow, triggering systemic hypoperfusion. A sudden severe drop in kidney perfusion can rapidly cause ischemia of the renal tubules, leading to acute tubular necrosis.³⁶ However, while low cardiac output initiates CRS pathophysiology, it is rarely the sole driver. A survey of the Acute Decompensated Heart Failure

National Registry (ADHERE) cohort demonstrated that while a vast majority (91%) of patients with ADHF presented to the hospital with renal dysfunction, <2% were hypotensive, with systolic blood pressure <90 mm Hg.³⁷ Further, in the ESCAPE trial (Evaluation Study of Congestive Heart Failure and Pulmonary Artery Catheterization Effectiveness), there was no association between cardiac index and AKI.⁷ In CS, higher cardiac output, not lower, is associated with worsening renal function,⁴ suggesting that CS-CRS may be driven in part by pathophysiologic or iatrogenic factors that increase cardiac output during the transition to a mixed shock state.

Prerenal hypoperfusion also activates the renin-angiotensin-aldosterone system (RAAS), which triggers sympathetic nervous system activation and arginine vasopressin secretion.³⁸ Ang II and aldosterone secretion then result in sodium reabsorption that expands circulatory volume, resulting in a vicious cycle in which expanded volume increases central venous pressure (CVP), worsens renal venous congestion, and further impedes intrarenal blood flow. The combination of systemic hypoperfusion and elevated CVP results in a decreased gradient from the renal artery to the venous system, reduced RPP, and a decrease in glomerular filtration rate.³⁹ Elevated CVP and venous congestion were the only hemodynamic predictors of renal dysfunction in ADHF in ESCAPE,⁴⁰ although these only explained a small (2%) portion of the AKI risk. Elevated right-sided and intraabdominal pressures may

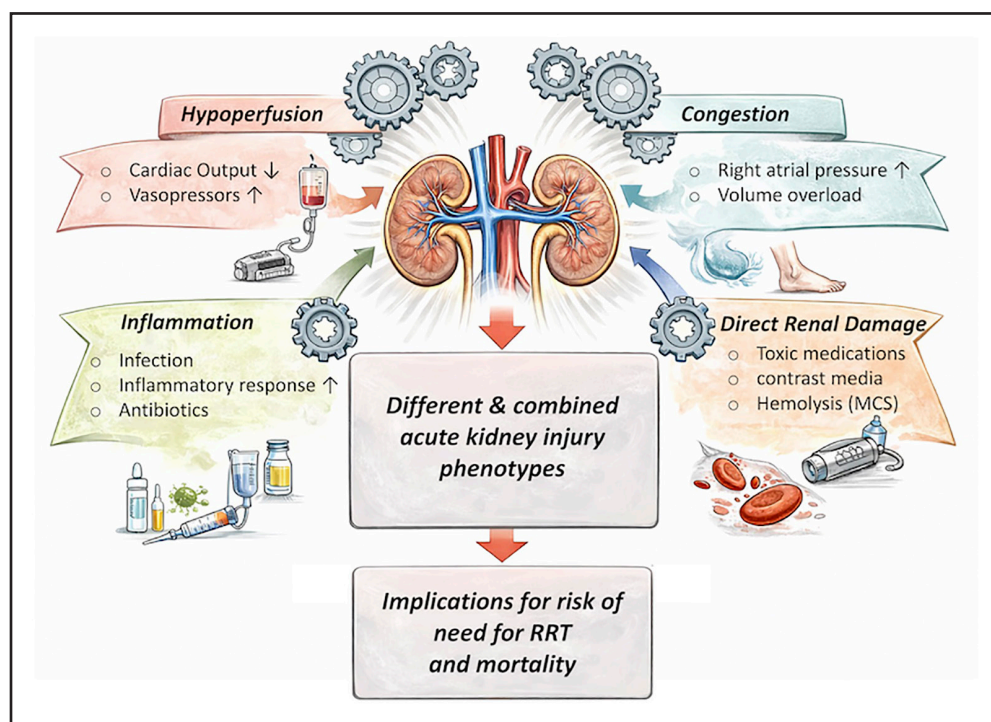


Figure 1. Pathophysiologic mechanisms in cardiorenal syndrome (CRS) in cardiogenic shock (CS).

CRS pathophysiology is heterogeneous and may differ by CS cause (eg, acute myocardial infarction vs acute heart failure vs acute on chronic heart failure), cardiometabolic phenotype, preexisting chronic kidney disease, iatrogenic exposures, and hemodynamic profile. Individuals with CRS require comprehensive evaluation to understand mechanisms of renal dysfunction and to tailor therapeutic interventions. MCS, mechanical circulatory support; and RRT, renal replacement therapy.

exacerbate renal dysfunction through compression of the renal parenchyma and diminished perfusion.⁴¹ Along with these hemodynamic changes, there is often concomitant osmotic dysregulation from vasopressin-mediated free water retention.⁴²

Right Heart Dysfunction

Right heart dysfunction impairs renal function through multiple concomitant pathways. In addition to its effects on renal venous pressures, right heart dysfunction also alters ventricular interdependence, reducing left ventricular filling and cardiac output. Echocardiographic evidence of right heart dysfunction has previously been shown to be associated with worsening renal outcomes.⁴³ Conversely, right heart dysfunction at ADHF onset may predict the ability of the kidneys to improve in response to successful decongestion,^{43,44} but data are scarce for CS.

Neurohormonal Maladaptation and Inflammation

The low-flow state of CS leads to RAAS activation and sympathetic overdrive as compensatory mechanisms to expand intravascular volume and maintain renal perfusion.⁴⁵ Ang II (angiotensin II)-mediated vasoconstriction also aids blood pressure homeostasis and causes efferent arteriolar vasoconstriction of the glomerulus, which can preserve glomerular filtration rate despite significant reductions in renal blood flow.⁶ However, if sustained for a prolonged period, RAAS effects become maladaptive and result in oxidative stress, inflammation, and fibrosis in both the heart and kidneys.⁴⁶ In chronic heart failure, increased Ang II production from RAAS activation activates transcription factors nuclear factor kappa light chain enhancer of activated B cells and activator protein 1, which lead to the production of pro inflammatory cytokines such as tumor necrosis factor alpha, interleukin 6, and interleukin 1-beta.⁴⁷ Translational studies have confirmed that increased systemic inflammation is associated with worsening renal function in patients with ADHF.⁴⁸

Conversely, the relationship among Ang II, CS, and AKI may be heterogeneous and could evolve across SCAI stages or change between the acute and chronic settings. In animal models that emulate the later-stage transition from pure CS to mixed cardiogenic and vasodilatory CS, hydrolysis of Ang II by circulating DPP3 (dipeptidyl peptidase III) results in vasodilation and a drop in glomerular filtration rate.⁴⁹ Elevated DPP3 levels are strongly associated with mortality in CS, and this appears to be a ubiquitous end-stage phenomenon of vasodilatory collapse across multiple shock states, including CS, septic shock, burn shock, and severe trauma.⁵⁰ Loss of Ang II-mediated systemic and efferent glomerular arteriolar vasoconstriction could possibly explain the prevalence of AKI in higher SCAI CS stages or the association

of AKI with mortality. These mechanisms are being studied to better understand their role in CRS and whether they could be therapeutically targeted to improve kidney function and CS survival.

Mitochondrial Dysfunction

Systemic inflammation, high sympathetic tone, and neurohormonal overactivation can all lead to mitochondrial dysfunction in both cardiomyocytes and renal tubular cells. This produces excess reactive oxygen species,⁵¹ resulting in progressive cardiac and renal damage, remodeling, and fibrosis that propagate systemic inflammation.⁵²

Iatrogenic Contributors

Iatrogenic factors may worsen renal function in CS and differ between AMI-CS and HF-CS. For example, the use of contrast media during coronary angiography and percutaneous interventions for AMI-CS can cause AKI. Contrast-induced nephropathy is a well-documented complication, especially among patients with preexisting renal dysfunction or acute hemodynamic instability.⁵³ In patients presenting with ADHF-CS and fluid overload, diuretics often cause a rise in serum creatinine, although small changes may often represent hemoconcentration rather than true AKI.⁵⁴ Moreover, in the longer term, adequate decongestion is associated with improved survival and fewer hospital admissions despite rising serum creatinine.⁵⁵ The impact of shock-specific therapies, such as tMCS or vasoactive medications, are discussed later. Noncardiac therapies in the critically ill patients may also contribute to AKI, including an association between aminoglycosides, antifungals, and antivirals with acute tubular necrosis, β -lactams or proton pump inhibitors with acute interstitial nephritis, and a particularly high association with AKI when piperacillin-tazobactam is combined with vancomycin.⁵⁶

KIDNEY FAILURE PHENOTYPES IN CS

AKI in CS is not 1 entity—timing, hemodynamics, and biology often point to different dominant mechanisms and thus potentially different priorities for strategies to foster renal protection.

Early Versus Late AKI

AKI timing is associated with prognosis and may indicate important differences in cause (Figure 2).^{3,4} Most AKI (80%) occurs within 48 hours of admission to the CICU,³ and most escalations to RRT occur within 72 hours. De novo CS (either new ADHF or AMI) increases the likelihood of worsening renal function within 72 hours.⁴ In AMI-CS, early AKI marked a higher-risk phenotype, with higher RAP, lactate, vasopressor needs, and mortality than late AKI, indicating the relationship of early AKI directly to CS pathophysiology.³ In contrast, late AKI

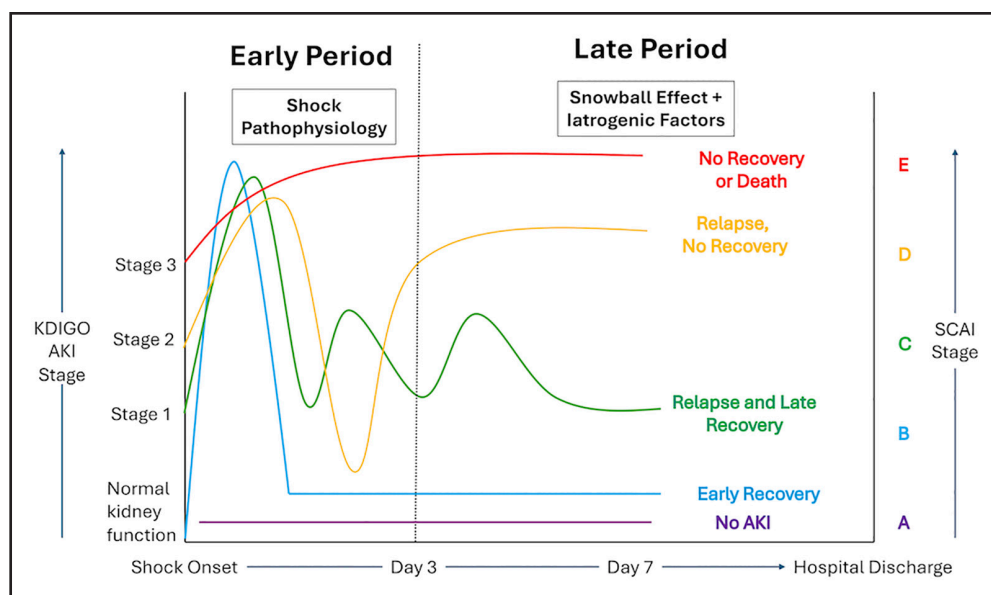


Figure 2. Temporal trajectories of kidney failure in cardiogenic shock.

Kidney function is often dynamic in cardiogenic shock (CS) and follows a handful of patterns with varying degrees of recovery. Early acute kidney injury (AKI) within 48 to 72 hours is more strongly associated with CS mortality than late AKI (>72 hours) and is driven by different mechanisms. The Kidney Disease: Improving Global Outcomes (KDIGO) stage at CS onset, as well as the Society for Cardiovascular Angiography (SCAI) CS severity stage, influence the pattern and likelihood of kidney recovery.

may accrue from various factors, including ongoing congestion, nephrotoxic exposures such as contrast media or antibiotics, or relate to tMCS, bleeding, inflammation, and infection,²⁶ and these etiologies of AKI are not as strongly associated with mortality.

AKI Versus AKI on CKD

AKI can arise de novo or with background CKD. With preexisting CKD, reduced renal and cardiac functional reserves contribute to accelerated bidirectional cardiac–renal dysfunction, exacerbated by endothelial dysfunction, oxidative stress, and systemic inflammation. Preexisting CKD is one of the strongest variables associated with AKI and alters the progression, prognosis, and management of CRS.⁵⁷ CKD prevalence in CS ranges from 12% to 34%, with increasing prevalence over time.^{4,31,58,59} In HF-CS, baseline CKD is associated with mortality, SCAI stage, lactate levels, sepsis, RRT, vasopressors, and tMCS, although notably did not affect outcomes in the tMCS group.²⁶ Patients with ESRD represent a unique subgroup with a rising incidence of CS and no improvement in mortality over time despite increasing tMCS utilization.⁵⁹

AKI in Cardiorenal Versus Cardiometabolic CS

Machine learning-derived CS clusters identified 3 groups with distinct associations among key metabolic and hemodynamic variables.^{16,60} Of these, the cardiometabolic cluster, followed by the cardiorenal cluster, has the

highest AKI risk and may represent more type 1 CRS.⁶¹ Instead, patients who cluster in the cardiorenal phenotype tended to be older, with lower heart rate, elevated pulmonary arterial and pulmonary capillary wedge pressures, a higher prevalence of CKD, and the lowest presenting glomerular filtration rate, suggesting a higher predominance of type 2 CRS. In the cardiometabolic cluster, patients exhibited a higher lactate, liver enzymes, heart rate, and right atrial pressure along with lower blood pressure, cardiac power output, and cardiac index. However, the mechanistic contributions of congestion, malperfusion, or other factors as the drivers of this mortality in these clusters remain unclear.

Additional Phenotypes

A subset of patients develops a systemic inflammatory response from CS itself or secondary sepsis that contributes to microcirculatory dysfunction and tubular injury—over and above macrocirculatory indices.⁶² In line with this, sepsis has recently been confirmed as a relevant contributor to AKI in patients with HF-CS.²⁶

Other phenotypes of AKI in CS may arise from bleeding events²⁶ and potentially concurring hemodynamic fluctuations, as well as hemolysis from sepsis or tMCS use. Specifically, the use of microaxial flow pumps in the DanGer Shock Trial was associated with a higher risk of AKI and RRT, potentially related to increased hemolysis.²⁴ However, none of the patients in the device arm required long-term dialysis, indicating this specific AKI phenotype is often reversible.

DIAGNOSTIC AND MONITORING STRATEGIES

Traditional Measures

The current clinical definition of AKI is based on KDIGO guidelines, with increases in serum creatinine $\geq 1.5\times$ baseline within 7 days or ≥ 0.3 mg/dL within 48 hours, or a decrease in urine output <0.5 mL/kg per hour for 6 hours, as the standard criteria (Table 1).¹⁷ AKI is further classified into 3 progressive stages based on the severity. Once AKI is established, a careful clinical history, physical examination including objective assessment of volume status, laboratory evaluation, and imaging studies are recommended (Table 3) to evaluate for causes of AKI. Additional serological workup and kidney biopsy may be needed for suspicion of amyloidosis, glomerulonephritis, or other non-CRS AKI etiologies. Limitations of the KDIGO classification in CS are discussed above.

Hemodynamic Assessment

Hemodynamic parameters such as mean arterial pressure, CVP and RPP are important to monitor in CS to diagnose or prevent AKI. Elevated CVP >10 mmHg has been consistently shown to be an independent predictor of AKI development⁵⁷ whereas mean arterial pressure <65 mmHg, despite its association with worse outcomes, does not reliably predict AKI.⁶³ RPP, defined as the difference between mean arterial pressure and CVP, has been shown to predict the risk of new-onset and progression of AKI.^{63,64} The role of RPP in CS has not been fully ascertained, and beneficial cutoffs not well established, but maintaining an RPP >60 mmHg is prudent and potentially higher in patients with chronic hypertension or CKD.

Pulmonary artery catheterization helps differentiate between right, left, or biventricular CS, which have distinct implications for AKI pathophysiology and management. Of the various parameters, reduced central venous oxygen saturation is independently associated with AKI development and mortality and has prognostic value in monitoring response to decongestive measures.⁶⁵ However, it is important to note that pulmonary artery catheterization provides hemodynamic context for interpreting AKI rather than diagnosing AKI directly. Results from ongoing clinical trials such as the NorShock Trial (REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05168462) and Pulmonary Artery Catheter in Cardiogenic Shock (PACCS, REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05485376) will provide much-needed insight into the role of pulmonary artery catheterization guided management.

Emerging Tools and Biomarkers

Several novel biomarkers have been investigated to predict AKI and refine prognosis in CRS (Table 3). These

may better capture renal injury phenotypes, particularly those related to congestion, endothelial dysfunction, or early structural damage. These include clearance markers (cystatin C), tubular injury markers (NGAL [neutrophil gelatinase-associated lipocalin], KIM-1 [kidney injury marker-1], L-FABP [liver-type fatty acid-binding protein], interleukin-18), cell-cycle arrest markers (TIMP-2 [tissue inhibitor of metalloproteinases-2] and IGFBP7 [insulin-like growth factor-binding protein 7]), and other neurohumoral markers (angiopoietin, angiotensinogen). Among these, urinary TIMP2-IGFBP7 and NGAL have the strongest evidence for predicting AKI, with the former currently Food and Drug Administration-approved for detecting moderate-to-severe AKI.^{66,67} Although both biomarkers have area under the curve close to 0.8 for AKI, the long-term impact on patient outcomes remains uncertain.⁶⁶ Importantly, these biomarkers help distinguish hemodynamic fluctuations in kidney function from true tubular injury.

Renal Doppler ultrasound offers a noninvasive assessment of renal vascular resistance, CKD prognosis, and venous congestion⁶⁸ that could help to distinguish type 1 and type 2 CRS. The renal resistive index, calculated from interlobar renal arteries, measures intrarenal vascular resistance and reflects fibrosis, CKD risk, and renal congestion, whereas the intrarenal venous flow patterns, obtained from interlobar renal veins and classified as continuous (normal), biphasic, or monophasic, are associated with severe venous congestion.⁶⁹ The venous excess ultrasound protocol integrates assessment of IVC diameter with pulsed-wave Doppler evaluation of the hepatic, portal, or renal veins to grade systemic congestion, with higher grades predictive of reduced diuretic efficiency and in-hospital mortality⁷⁰ in CRS. Although drawbacks include large interindividual variability and operator dependence, Doppler ultrasound in combination with hemodynamic monitoring may provide valuable information to guide treatment.

IMPACT OF CS THERAPIES ON THE KIDNEY

Pharmacological Support

Vasopressors

Limited data are available concerning the effects of vasopressor therapy on AKI risk in CS. Evidence from septic shock suggests that high catecholamine doses correlate with increased AKI incidence and severity.⁷¹ This might be attributable to renal vasoconstriction and microcirculatory dysfunction but could also reflect more severe shock. A causal link remains unconfirmed. Like the effects on all-cause mortality in CS, the effects of vasopressor use on AKI risk in CS in general remains uncertain. However, some data are available comparing different vasoactive agents. In the OPTIMA-CC

Table 3. Diagnostic Tools for AKI in CS

Diagnostic tool	Key features/Utility	Limitations
History	Medications specifically NSAIDs, ACE inhibitors/ARB, diuretics are helpful in assessing AKI causes	
Physical examination	Assess volume status and perfusion	Subjective assessment dependent on examiner's expertise
SCAI shock classification	Standardized tool that helps in treatment decisions and predicts mortality risk	Not designed to diagnose AKI
Hemodynamic monitoring		
Central venous pressure	>10 mm of Hg is predictive of AKI and has prognostic value	Lowering CVP does not reduce incidence or severity of AKI
Mean arterial pressure	<65 mmHg is associated with worse outcomes	Does not predict AKI
Renal perfusion pressure	Low RPP (<60 mmHg) predicts risk of new-onset and progression of AKI	Not validated in CS
Central venous oxygen saturation (SvO ₂)	Independently associated with AKI development and mortality	Not a direct way to diagnose AKI
Traditional laboratory measures		
Serum creatinine	Defines AKI stages per KDIGO guidelines; baseline marker for kidney function	Affected by muscle mass and is not altered until significant kidney injury has occurred
Blood urine nitrogen (BUN)	Marker of renal hypoperfusion; baseline marker for kidney function	Nonspecific biomarker, cannot differentiate from other causes of kidney injury
Urine output	Defines AKI stages per KDIGO guidelines	Inconsistent measurement, is not altered until significant kidney injury has occurred
Lactate	Marker of tissue hypoperfusion	Nonspecific and does not help in AKI diagnosis
Urine analysis and microscopy	Diagnostic of intrinsic AKI, especially glomerulonephritis if there is an active sediment with casts or cells	Benign urine sediment is common with hemodynamic changes and cannot differentiate between prerenal causes of AKI
Fractional excretion of sodium and urea	Distinguishes prerenal AKI from intrinsic causes	Cannot differentiate between cardiorenal vs prerenal
BNP/NT-proBNP	Marker of congestion	Elevated in all CRS subtypes
Emerging biomarkers		
Serum cystatin C	Higher sensitivity for AKI than creatinine and not affected by muscle mass	Altered by changes in WBC counts such as sepsis
NGAL (neutrophil gelatinase-associated lipocalin)	High urine and plasma values predict AKI, need for dialysis and death	Time and context dependent and lack of standardization across assays
Urinary tissue inhibitor of metalloproteinases-2 (TIMP-2) and insulin-like growth factor-binding protein 7 (IGFBP7)	Cell-cycle arrest markers, FDA-approved to predict moderate-to-severe AKI	Time and context dependent
KIM-1 (kidney injury molecule 1)	Plasma and urine levels predict tubular injury in response to ischemia/toxins	Does not help in CS
L-FABP (liver-type fatty acid-binding protein)	Urinary levels can predict AKI before creatinine increase	Affected by liver dysfunction and CKD and reflects hypoperfusion-mediated injury
Interleukin-18	Strong predictor of AKI progression in early AKI and reflects inflammation	Less specific than NGAL and peaks later than other biomarkers
Imaging		
Kidney ultrasound	Evaluates kidney size, echogenicity and excludes obstructive causes of AKI	Cannot distinguish prerenal vs intrinsic AKI
Echocardiogram	Assesses biventricular and valvular function, mechanical complications, filling pressures, and forward flow	Not helpful for diagnosis of AKI directly
Renal Doppler	Venous excess ultrasound (VExUS) predictive of reduced diuretic efficiency and mortality; intrarenal venous flow (IVRF) predictive of venous congestion	Large interindividual variability and operator dependent

Multiple tools exist to aid the diagnosis of AKI in CS. Traditional historical- and laboratory-based diagnostic criteria can be supplemented by hemodynamic measures, imaging measures, and novel laboratory measures to better understand the most important cause and pathophysiologic mechanisms contributing to AKI in an individual patient and to separate true kidney injury from benign changes to glomerular filtration. ACE indicates angiotensin-converting enzyme; AKI, acute kidney injury; ARB, angiotensin receptor blocker; BNP, B-type natriuretic peptide; CKD, chronic kidney disease; CRS, cardiorenal syndrome; CS, cardiogenic shock; FDA, Food and Drug Administration; KDIGO, Kidney Disease: Improving Global Outcomes; NGAL, neutrophil gelatinase-associated lipocalin; NSAID, nonsteroidal anti-inflammatory drug; NT-proBNP, N-terminal pro B-type natriuretic peptide; RPP, renal perfusion pressure; SCAI, Society for Cardiovascular Angiography and Interventions; and WBC, white blood cell.

Trial (Optimizing the Use of Vasopressor After Coronary Reperfusion in Cardiogenic Shock Secondary to Myocardial Infarction),⁷² epinephrine use was associated with a higher tendency for the need for RRT compared with norepinephrine (26% versus 7%; $P=0.07$). Similarly, in the SOAP II trial (Sepsis Occurrence in Acutely Ill Patients II),⁷³ norepinephrine was associated with more days free of RRT compared with dopamine (14 versus 12.8 days; $P=0.07$). No trials have compared norepinephrine with vasopressin in CS, except in postcardiotomy shock, where vasopressin might be advantageous based on the small VaNCS study (Vasopressin versus Norepinephrine in Patients with Vasoplegic Shock After Cardiac Surgery).⁷⁴ In septic shock, RRT risk was similar with both agents in the Vasopressin and Septic Shock Trial (VASST),⁷⁵ but lower with vasopressin in the smaller VANISH trial (Vasopressin versus Noradrenaline as Initial Therapy in Septic Shock).⁷⁶ A post hoc analysis of the Angiotensin II for the Treatment of High-Output Shock (ATHOS-3) trial in distributive shock showed that patients randomized to Ang II had greater liberation from RRT at 7 days than those receiving placebo,⁷⁷ perhaps related to its preferential constriction of efferent glomerular arterioles compared with catecholamines, and there may be subsets of patients with CS or mixed shock in whom this applies. However, more evidence is required to guide decision-making with respect to tailored vasopressor therapy in CS.

Inotropes

Similarly, the evidence with respect to the choice of inotrope therapy is scarce. The most used inotropic agents in CS are dobutamine, phosphodiesterase III inhibitors such as milrinone or enoximone, and levosimendan.⁷⁸ Norepinephrine, epinephrine, and dopamine, which also have inotropic effects, are discussed above. Although the concept of improving cardiac output with inotropes to improve renal perfusion is plausible, there are no data clearly supporting the benefit of individual strategies. In the DOREMI trial (Milrinone as Compared with Dobutamine in the Treatment of Cardiogenic Shock), milrinone treatment was associated with a hazard ratio of 1.39 (95% CI, 0.73–2.67) for RRT.⁷⁹ After early signals of renal benefits of levosimendan over dobutamine in the small Levosimendan Infusion versus Dobutamine (LIDO) study,⁸⁰ the larger Survival of Patients With Acute Heart Failure in Need of Intravenous Inotropic Support (SURVIVE) trial, in which dobutamine was compared with levosimendan in patients with ADHF, detected no differences in AKI between the treatment arms.⁸¹

tMCS

IABP

Although there is a rationale to stabilize blood pressure and coronary perfusion during CS, device-related

complications such as bleeding or renal artery occlusion may pose a risk to the kidneys. No study has yet proven a renal benefit of IABP utilization. The IABP-SHOCK-II trial (Intraaortic Balloon Pump in Cardiogenic Shock II) in AMI-CS reported a risk ratio of 1.31 for RRT in the IABP arm.⁹ The recent Study on Early Intra-aortic Balloon Pump Placement in Acute Decompensated Heart Failure Complicated by Cardiogenic Shock (ALTSHOCK-II) study concluded early and, with RRT used in just 4 patients, was underpowered to draw any conclusions for HF-CS.⁸²

VA-ECMO

Similarly, no study has proven a clear renal benefit of Extracorporeal Membrane Oxygenation (ECMO) use in CS. However, the risk of requiring RRT was lower in the ECMO arm of the ECLS-SHOCK trial (Extracorporeal Life Support in Infarct-Related Cardiogenic Shock; risk ratio, 0.58) despite an increased bleeding risk.⁸ This lower rate of RRT did not translate into improved mortality in ECLS-SHOCK.

Impella

The largest trial assessing renal outcomes in Impella-supported patients with CS is the DanGer Shock trial. In comparison with the IABP-SHOCK II and ECLS-SHOCK trials mentioned above, the overall risk of AKI was higher in the DanGer Shock trial population (proportion of patients in the control groups requiring RRT: IABP-SHOCK II, 15.7%; ECLS-SHOCK, 13.9%; DanGer Shock, 26.7%).¹⁰ In the DanGer Shock trial, Impella CP use was associated with a higher risk of AKI and RRT, even when accounting for the competing risk of death (subdistribution hazard for RRT, 1.67 [95% CI, 1.18–2.35]).²⁴ Predictors of AKI specific to Impella support were suction events, higher pump speed, and longer duration of support.²⁴ While hemolysis was not systematically assessed in DanGer Shock, surrogate markers suggested hemolysis as a major driver of the increased AKI risk in this cohort. At the same time, some explorative analyses have suggested a potential renoprotective mechanism of Impella support in high-risk PCI.^{83–85} Therefore, the exact mechanisms of AKI associated with the Impella CP remain unclear. Comparatively little data are available about the risk of AKI with larger microaxial flow devices such as the Impella 5.5. It is plausible, but not well established, that higher flow and less hemolysis would mitigate AKI risk.

Timing of tMCS Initiation and Renal Recovery

While data are limited, observational studies suggest that earlier initiation of tMCS, specifically ECMO may be associated with improved outcomes, including trends toward better renal perfusion and reduced AKI severity.⁸⁶ However, data are inconsistent,⁸⁷ and more research is needed to identify the optimal timing of tMCS application for renal protection in CS.

Fluid Management

Resuscitation Versus Diuresis

Fluid resuscitation in CS is a double-edged sword. Early fluid boluses may transiently increase preload and improve cardiac output, but excessive fluid leads to congestion, elevated renal venous pressures, and worsened AKI. Conversely, judicious fluid removal (with diuretics or ultrafiltration when appropriate) can reduce congestion and improve renal perfusion but must be carefully balanced to avoid excess intravascular volume depletion and tMCS malfunction, eg, due to suction events. According to data from the DanGer Shock trial, positive fluid balance within the first 24 hours was a strong predictor of AKI risk, but the direction of the causal effect is unclear. Conversely, negative fluid balance did not translate to lower AKI risk.²⁴ The DOSE study (Diuretic Strategies in Patients with Acute Decompensated Heart Failure) showed no differential benefit between bolus versus continuous drip diuretics, but this pertained primarily to patients with ADHF and not necessarily CS.⁸⁸

Dynamic metrics (eg, passive leg raise, CVP, echocardiographic assessments) help tailor resuscitation and decongestion to support renal recovery.⁷⁸ Particularly, venous excess ultrasound and left ventricular outflow tract velocity-time integral represent promising noninvasive approaches for targeted fluid management to achieve euvolemia.⁸⁹

RRT IN CS

RRT Target Population

The indications for RRT in CS mirror those for other critically ill patients with AKI—severe metabolic acidosis, electrolyte derangements such as hyperkalemia, intoxications, volume overload, and uremia.⁹⁰ The KDIGO guidelines recommend initiation of RRT for stage 2 AKI when life-threatening changes in fluid, electrolyte, and acid-base balance exist. In CS, severe pulmonary edema and clinically significant hyperkalemia with cardiac conduction abnormalities often require RRT. Of note, data from the FRENDSOCK registry (French Observatory on the Management of Cardiogenic Shock) found that the indication for RRT was associated with outcomes; patients dialyzed for metabolic disturbances (acidosis/hyperkalemia) had significantly higher 30-day mortality compared with those treated for fluid overload/oliguria (hazard ratio, 1.99 [95% CI, 1.08–2.68]; $P=0.02$).³⁴ The same study also demonstrated that elevated bilirubin, severe anemia, advanced CKD, infectious triggers, epinephrine use, mechanical ventilation, and mechanical circulatory support were independent predictors of RRT requirement in CS. A furosemide stress test evaluating urine output above or below 200 mL within 2 hours of a moderate loop diuretic predicted subsequent need for RRT, although earlier use of RRT in patients with

an inadequate urine response was not associated with improved 28-day survival or fluid balance.⁹¹

RRT Timing

The optimal timing of RRT initiation in CS remains unclear as most randomized timing trials have excluded patients with CS,^{90,92} although some inferences can be made from other critically ill or heart failure populations and from observational studies in CS. The AKIKI (Artificial Kidney Initiation in Kidney Injury Trial) and the larger STARRT-AKI trial (Standard versus Accelerated Initiation of Renal-Replacement Therapy in Acute Kidney Injury) did not show any survival benefit of RRT initiation in ICU patients within 12 hours of KDIGO stage 2 or 3 onset compared with standard care, although CS was not well represented.^{92,93} By comparison, the smaller ELAIN trial (Early versus Late Initiation of Renal Replacement Therapy in Critically Ill Patients with Acute Kidney Injury) suggested a possible benefit of early RRT in patients with KDIGO stage 2 AKI and biomarker evidence of renal tubular or parenchymal injury (NGAL levels >150 ng/mL).⁹⁴ In CARESS-HF (Cardiorenal Rescue Study in Acute Decompensated Heart Failure), patients with ADHF and worsening renal function had greater creatinine increase and adverse events with fixed-rate ultrafiltration compared with diuresis, underscoring the importance of individualized fluid-removal strategies and concomitant hemodynamic optimization.⁹⁵ A single-center study of patients with postcardiotomy CS found that preemptive RRT was associated with lower hospital mortality and faster and more frequent renal recovery compared with conventional indications.⁹⁶ In contrast, data from the FRENDSOCK registry showed no mortality difference between early versus late (<48 hours versus >48 hours) RRT initiation.³⁴ Similarly, a recent meta-analysis found no significant differences in 30-, 60-, or 90-day mortality, length of CICU or hospital stay, dependence on dialysis, or renal recovery between early and late strategies, although the definitions of timing varied widely across studies.⁹⁷

RRT Modality

RRT in CS can be delivered as continuous RRT (CRRT) over 24 hours, or intermittent hemodialysis (IHD) for 3 to 6 hours daily or on alternate days. CRRT is generally preferred over IHD in CS due to hemodynamic instability and intolerance of large intravascular volume shifts.⁹⁰ On the other hand, IHD may be advantageous for rapid fluid and solute removal in acute pulmonary edema or severe electrolyte imbalances and offers improved patient mobility, lower nursing workload, and reduced costs.⁹⁰ Hybrid RRT modalities such as sustained low-efficiency dialysis or prolonged intermittent RRT provide intermediate-duration therapy (8–16 hours) with fewer

resource demands (Figure 3). For patients on ECMO, CRRT is favored as it can be integrated into the circuit. The decision to initiate RRT with CRRT, IHD, or prolonged intermittent RRT is largely resource- and expertise-dependent. Randomized AKI trials show no clear difference in survival, renal recovery, or hemodynamic tolerance between modalities,⁹⁰ although large CS-specific trials are lacking.

An important consideration in RRT is the use of anticoagulant therapy, which is often needed to maintain circuit patency. Although this is easier to accomplish in patients on ECMO, other patients may require new use of heparin, increasing bleeding risk. Regional citrate anticoagulation is an alternative to heparin but requires close monitoring to prevent citrate toxicity and calcium imbalance. On the other hand, if anticoagulation is contraindicated, circuit

clotting and treatment interruption could affect RRT efficiency.

Outcomes Associated With RRT

RRT requirement in CS is associated with substantially worse outcomes.⁹⁸ In a propensity-matched analysis, patients with CS requiring RRT had significantly higher in-hospital mortality, longer lengths of stay, and higher costs.⁵ Further, there is high variability (between 20% and 60%) in the discontinuation of CRRT among CICU patients. While urine output of >400 mL/24 hours without diuretic use is the strongest predictor for discontinuation of CRRT,⁹⁹ up to 33% of patients on RRT may need to be transitioned to IHD until renal recovery. Transition to IHD depends on hemodynamic stability

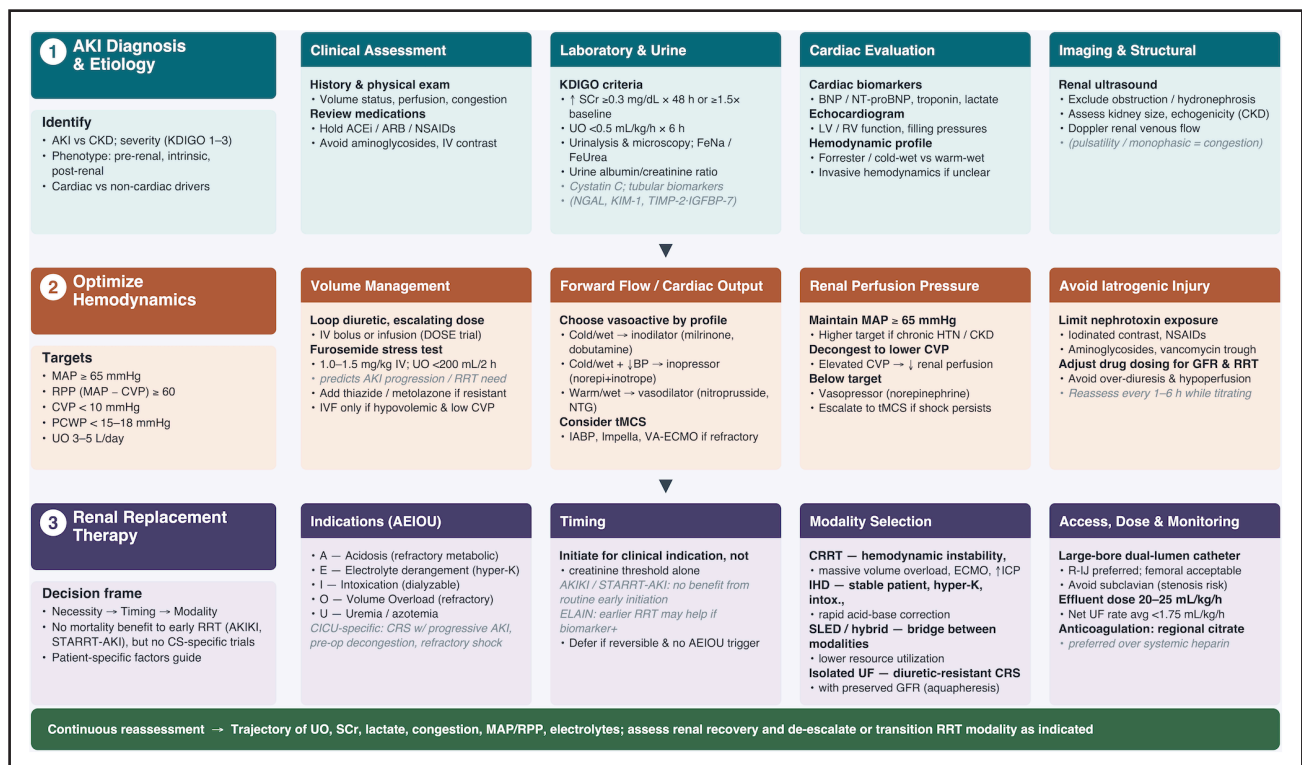


Figure 3. Stepwise diagnostic and management algorithm for acute kidney injury (AKI) in cardiogenic shock.

Step 1 establishes the diagnosis and cause of AKI through clinical, laboratory, imaging, and hemodynamic assessment. Step 2 optimizes systemic and renal hemodynamics by guiding volume management, cardiac output, renal perfusion pressure, and the avoidance of nephrotoxic agents.⁹¹ Step 3 addresses renal replacement therapy (RRT) indications, timing, modality, and access. These steps are iterative and require ongoing reassessment and management. ACE indicates angiotensin-converting enzyme; AKIKI, Artificial Kidney Initiation in Kidney Injury;⁹³ ARB, angiotensin receptor blocker; BNP, B-type natriuretic peptide; BP, blood pressure; CARRESS-HF, Cardiorenal Rescue Study in Acute Decompensated Heart Failure;⁹⁵ CICU, cardiac intensive care unit; CKD, chronic kidney disease; CRRT, continuous renal replacement therapy; CRS, cardiorenal syndrome; CVP, central venous pressure; DOSE, Diuretic Optimization Strategies Evaluation;⁸⁸ ECMO, extracorporeal membrane oxygenation; ELAIN, Early vs Late Initiation of Renal Replacement Therapy in Critically Ill Patients with AKI;⁹⁴ FeNa, fractional excretion of sodium; FeUrea, fractional excretion of urea; GFR, glomerular filtration rate; HTN, hypertension; IABP, intraaortic balloon pump; ICP, intracranial pressure; IGFBP7, insulin-like growth factor-binding protein 7; IHD, intermittent hemodialysis; IV, intravenous; IVF, intravenous fluids; KDIGO, Kidney Disease: Improving Global Outcomes; KIM-1, kidney injury molecule 1; LV, left ventricular; MAP, mean arterial pressure; MCS, mechanical circulatory support; NGAL, neutrophil gelatinase-associated lipocalin; NSAID, nonsteroidal anti-inflammatory drug; NTG, nitroglycerin; NT-proBNP, N-terminal pro-B-type natriuretic peptide; PCWP, pulmonary capillary wedge pressure; R-J, right internal jugular vein; RPP, renal perfusion pressure; RV, right ventricular; SCr, serum creatinine; SLED, sustained low-efficiency dialysis; STARTR-AKI, Standard vs Accelerated Initiation of Renal Replacement Therapy in AKI;⁹² TIMP-2, tissue inhibitor of metalloproteinase 2; tMCS, temporary mechanical circulatory support; UF, ultrafiltration; UO, urine output; and VA-ECMO, venoarterial extracorporeal membrane oxygenation.

with minimal vasopressor support and tolerability of fluid removal with IHD on a daily or alternate-day basis. In general, dialysis dependence after day 90 is associated with lower renal recovery and defines end-stage CKD. A Danish population-based cohort study found that among hospital survivors, the 5-year risk of chronic dialysis was 11% for patients requiring RRT versus 1% for those not requiring RRT, with 5-year mortality of 43% in the RRT group versus 29% in the non-RRT group.²³ Overall, both short- and long-term morbidity and mortality remain high in patients with CS requiring RRT, underscoring the need for improved diagnostic and therapeutic strategies.

NOVEL CARDIORENAL DEVICES

Specific renal assist devices have been developed to circumvent the cycle of progressive hemodynamic and neurohormonal perturbations that characterize CRS (Table 4), but with fewer potential harmful effects and more out-of-hospital support potential than existing tMCS. These devices aim to improve the pressure gradient across the kidneys by either increasing renal artery

perfusion pressure, termed pushers, or by decreasing renal venous hypertension, termed pullers. Pushers are inserted via the femoral artery, vary in regard to sheath size (10F to 22F), and all involve active pump function via various mechanisms to generate the push. Current pushers being investigated are further detailed in Table 4. Among these, the Aortix (Procyron), an axial flow pump positioned superior to the renal arteries, is the furthest along in development. It is reported to augment cardiac output by 3.5 L as well as decrease left ventricular afterload. Multicenter pilot data has demonstrated feasibility, with a hypothesis-generating signal toward improved diuresis and hemodynamics. A larger, multicenter randomized trial, Diuretics alone versus Aortix Endovascular Device for Acute Heart Failure (DRAIN-HF, REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05677100) is currently ongoing. The remaining 3 pushers—the Reitan Catheter Pump (Cardiobridge), Second Heart Assist (Second Heart Assist, Inc), and ModulHeart (Puzzle Medical Devices)—remain in early developmental stages.

In contrast, pullers are accessed via the venous system (internal jugular or femoral vein) and have a

Table 4. Devices in Development for Cardiorenal Syndrome

Device class	Device name	Manufacturer	Description and mechanism	Access and location	Development stage
Pusher	Reitan catheter	Cardiobridge	Intraaortic propeller pump with foldable pump head. Aims to increase renal perfusion and decrease cardiac afterload	10F femoral arterial access to descending aorta	Early feasibility
	Aortix	Procyron	Percutaneous axial-flow aortic pump. Aims to increase renal perfusion and decrease cardiac afterload	18F inner sheath/21F external sheath femoral arterial access to descending aorta	Investigational Device Exemption trial
	ModulHeart	Puzzle Medical Devices	Modular multipump endovascular system with 3 smaller pumps in parallel. Aims to increase renal perfusion and decrease cardiac afterload	22F femoral arterial access to descending aorta	Early feasibility
	Second Heart Assist	Second Heart Assist	Aortic stent-mounted axial pump. Aims to increase renal perfusion and decrease cardiac afterload	13.5F femoral arterial access into descending aorta	Preclinical/ Early Human transition
Puller	Doraya Catheter	Revamp Medical	Passive IVC flow regulator to partially obstruct blood below renal veins	IVC below renal veins via femoral access	Early feasibility
	Precardia	Johnson and Johnson	Intermittent balloon-based SVC occlusion system	SVC via internal jugular	Early feasibility
	Perfuser	Nephronyx	Passive venous flow accelerator to draw renal venous blood into IVC via Bernoulli effect	IVC near renal veins via femoral vein	Early feasibility
	Magenta Transcatheter Renal Venous Decongestion System	Magenta Medical	Axial flow pump positioned near renal veins to actively aid decongestion	Infrarenal vena cava via femoral vein	Early feasibility
Fluid Shifter	WhiteSwell system	Whiteswell	Lymphatic drainage catheter	Thoracic duct outflow via left internal jugular	Early feasibility
	AquaPass	Aqua-Aerobic Systems	External thermal suit inducing sweat via increased skin temperature	Skin suit	Early feasibility
	Reprieve	Reprieve Cardiovascular	Automated diuretic management system that titrates diuretics and fluid replacement based on urine output	Pharmacological	Pivotal Trial
	Direct Sodium Removal (DSR) 2.0	Sequana Medical	Direct sodium removal via infusion of sodium free fluid to peritoneal cavity	Peritoneal cavity	Phase 1/2a

Several novel pushers, pullers, and fluid shifters are being studied in cardiorenal syndrome that could be applicable to patients with cardiogenic shock. These vary by access, mechanism, target physiology, and development phase. F indicates French; IVC, inferior vena cava; and SVC, superior vena cava.

more varied approach to their hemodynamic mechanisms. Current pullers under investigation include the Precardia system (Johnson and Johnson), a catheter-mounted balloon sitting in the superior vena cava that decreases filling pressures by intermittent balloon inflation resulting in transient superior vena cava occlusion and shown to be well tolerated in an early feasibility study;¹⁰⁰ the Magenta Transcatheter Renal Venous Decongestion System (Magenta Medical, REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT03621436), an inferior vena cava axial flow active pump positioned in the infrarenal vena cava; and 2 devices with passive mechanisms of action—the Doraya Catheter (Revamp Medical, REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05206422) and the Nephronyx system (Nephronyx, Ltd, REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT05759806) that both sit in the infrarenal vena cava and attempt to decrease renal venous pressures. A third category of fluid-shifter devices in development directly target fluid removal via targets such as the lymphatic system, interstitium, and microcirculatory plasma capillary refill rather than targeting renal hemodynamics.

The advent of these devices signals a willingness to explore a shift in decongestive approaches from a pure reliance on diuretics to direct targeting of abnormal renal hemodynamics or nontraditional direct fluid-removal strategies. However, these devices need to demonstrate clinical efficacy and acceptable device safety profiles. Clinical investigators will need to focus on identifying optimal target populations and clinical trial end points. Prior trials have focused on diuretic resistance, but this approach has logistic, regulatory, and physiological deficiencies. Several putative benefits of minimizing diuretic use exist, including the reduction in neurohormonal activation, which could reverse sodium avidity, adverse myocardial remodeling, and arrhythmias. Definitions of diuretic resistance, as well as classification of true kidney dysfunction versus benign serum creatinine changes remain inconsistently defined. Optimal clinical trial end points will need to balance showing clinical efficacy with trial feasibility and cost. Trial end points will thus likely need to combine hard end points such as mortality and readmissions with softer end points such as net fluid loss, changes in kidney function (perhaps using novel biomarkers such as NGAL), patient-reported outcomes, and device complications. Cost-effectiveness analyses that incorporate direct and indirect device costs as well as potential effects on length of stay, also need investigation. Devices have changed the paradigm of treatment for advanced heart failure and CS, but the role of these novel pushers, pullers, and other devices in ADHF and CRS remains to be seen.

CRS IN SPECIAL POPULATIONS

Postcardiotomy CS

AKI after cardiac surgery affects 12% to 50% of patients, with the incidence varying by type of surgery, and it adversely impacts outcomes.^{101–104} A smaller percentage of surgical patients, 2% to 6%, go on to require RRT, which notably worsens prognosis further.¹⁰⁵ Shock affects 2% to 6% of postcardiac surgery patients and places patients at higher risk of AKI.^{105,106}

The pathophysiology of AKI after cardiac surgery is multifactorial and incompletely understood. In addition to the hemodynamic insult of cardiac surgery with resultant periods of hypotension and possible acute tubular necrosis, multiple additional putative mechanisms include ischemia/reperfusion, nephrotoxin exposure, inflammation, oxidative stress, and microembolism, as well as iron-mediated toxicity and neurohormonal dysregulation such as acute Ang II depletion.¹⁰⁷ Longer duration of cardiopulmonary bypass increases AKI risk, potentially through decreased pulmonary conversion of Ang I to Ang II and increased systemic inflammation. Preexisting CKD remains the strongest risk factor for postoperative AKI and amplifies the association observed with cardiopulmonary bypass duration.^{107–109} Multiple other variables encompassing presurgical comorbidities, perioperative factors, and postoperative events are associated with AKI, including older age, preexisting heart failure or left ventricular dysfunction, prolonged ventilation, blood transfusions, and the presence of comorbidities, among others.¹¹⁰ Development of AKI after cardiac surgery is associated with increased short- and long-term mortality, stroke, kidney outcomes, increased CICU and non-CICU length of stays and health care costs.^{104–106} While both duration and severity of AKI amplify the negative impact on prognosis, even mild AKI demonstrates an association with worse outcomes.¹¹⁰ Societal guidelines and consensus statements emphasize early recognition, judicious cardiopulmonary bypass management, and careful hemodynamic monitoring in the perioperative period to mitigate CRS risk.^{103,109} Novel strategies such as extracorporeal blood purification, among others, are also being studied for efficacy in decreasing AKI risk after cardiac surgery.¹¹¹

OHT/LVAD

Similar to data in other populations, presence of CKD or AKI before or after durable left ventricular assist device or heart transplantation portends a worse prognosis.^{112–114}

Distinguishing AKI that could potentially reverse with improved cardiac support from permanent intrinsic kidney disease is pivotal to risk stratification and establishing left ventricular assist device or heart transplantation candidacy, but this can be difficult.¹¹⁴ AKI affects treatment

options in this acute stage of CS as it may limit exit strategies from tMCS. While CKD and even ESRD are not absolute contraindications to left ventricular assist device implantation, they profoundly affect mortality and quality of life. Large registry data has identified the combination of AKI and CS as carrying the highest mortality postleft ventricular assist device although younger patients appear to have more sustained renal recovery.^{113,114} If patients are able to be sufficiently stabilized, discussion regarding combined heart kidney versus safety-net kidney are further fraught with uncertainty regarding renal recovery, the increased risk of transplantation in an acutely ill patient, and the increased risk of heart alone transplantation in the face of CKD.¹¹⁴ If patients are successfully stabilized, the presence of CRS, whether acute or chronic, can further affect GDMT options and tolerability, thereby further affecting clinical trajectory. Collectively, this evidence, albeit observational in nature, underscores the role of CRS as a key determinant of outcomes in patients with CS.

CONCLUSIONS AND FUTURE DIRECTIONS

Although much progress has been made in our understanding of CRS and its association with and impact on CS, several fundamental questions still exist that must be overcome to advance the field.

First, while we have gained significant insight into the varying pathophysiological mechanisms that may be at play in CRS resulting in AKI, we need to be able to better recognize which mechanisms are at play in which patients. This is a fundamental step toward our ability to develop and tailor specific cardiorenal therapies for individuals with CS, including traditional therapies such as vasoactive medications and tMCS and more novel therapies such as cardiorenal pushers and pullers. It is likely that each of these therapies has a heterogeneous treatment effect across the spectrum of CRS in CS, and we must strive to narrow down a personalized approach for the smallest and best-defined subgroups possible.

Second, we must not rest assured that we know every mechanism at play in CRS. Instead, there are novel and provocative concepts that we have begun to observe, such as microvascular dysfunction related to CS or vasoactive medications, or the transition of RAAS activation to a state of potential RAAS depletion or dysregulation observed in animal models,⁴⁹ that require further investigation in patients. Each of these may offer novel targeted therapeutic strategies, and there are likely many unknown unknowns in this space that require our continued humble reassessment of the facts.

Given the strong and consistent relationship between CRS and adverse CS outcomes, including mortality, our ability to understand and treat the complex interplay

between the heart and kidneys, and the unseen factors that link these 2 organs, may ultimately be key to advancing CS care.

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Disclosures

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